
Refinement Buys Intelligibility, Search Buys Identity: What Test-Time Compute Buys in Masked-Diffusion TTS

Nityanand Mathur*

Hamees Sayed**

Ayush Pratap Singh

 tensorViz

 nityanandmathur/aspect-d  nityanandmathur/aspect-d-masked-diffusion-tts

Abstract

A diffusion language model buys sequential computation twice: depth, paid once in parameters, and denoising steps, rented per utterance at serve time. We ask whether the rented kind helps every capability equally. We train an iso-parameter width–depth grid of 15 masked-diffusion codec TTS models (3 seeds, 19–133 M non-embedding parameters) on 2,000 h of Emilia-EN and sweep the refinement budget T at inference only, evaluating cross-sentence zero-shot synthesis on 400 items from 174 held-out speakers. Our central result is model-free and referenced to floors we *measure* rather than to zero: the word error of an ASR system on the real recordings, and the speaker similarity of a codec round trip. Against those floors, refinement from $T=1$ to $T=16$ closes 86.2% of the reachable intelligibility range but only 46.4% of the reachable identity range, a $1.86\times$ asymmetry that holds between $1.68\times$ and $1.86\times$ under every monotone reparameterisation of the error we tried. Retraining the same runs to $3\times$ and $6\times$ the schedule attenuates that factor monotonically ($1.84\times$, $1.36\times$, $1.23\times$) while never reversing it, because refinement’s intelligibility share saturates (-0.09 pp across the last doubling, and -0.06 pp under a second recogniser from a different family) while its identity share keeps rising by $+7.4$ pp. The two capabilities have different bottlenecks, and we therefore offer the ordering, not the factor, as the result. We also report the pre-registered power-law fit, and we report that the step-exponent contrast it yields ($\Delta\tau = \tau_{\text{WER}} - \tau_{\text{SIM}} = 0.110$) changes sign under those same reparameterisations: it is a property of the error coordinate, and we therefore decline to build the paper on it. The limit is on what *refinement* buys, not on what inference buys. At a matched number of function evaluations (NFE), best-of- K search moves speaker similarity where refinement cannot, and 4 speaker-encoder families outside the selector’s lineage agree, with per-item win rates of 64.6–79.0% and every interval above one half. Depth and steps are *not* interchangeable: the substitution form $B(dT^\kappa)^{-\beta}$ fits worse than the separable form by $\Delta\text{AICc} = +69.3$. Finally, 62% of the remaining identity gap belongs to the codec, not the model, so most of what identity has left to gain is not the generator’s to win.

1 Introduction

A diffusion language model can buy sequential computation twice. It *owns* depth: d transformer blocks, paid once in parameters and then on every forward pass forever. It *rents* refinement steps: T denoising passes over the same sequence, paid per utterance and adjustable at serve time. An autoregressive model has only the first lever; its serial cost per token is fixed by the architecture the

*Work done outside of Blackstar Inc.

**Work done at Smallest AI.

moment training ends. If rented serial compute substitutes for owned serial compute, a deployment can trade parameters for steps after the fact. That trade is available only for the capabilities that steps improve.

Zero-shot text-to-speech is a clean probe: intelligibility is compositional (the right phonetic content in the right order, read out by word error rate), while speaker identity is representational (timbre carried from a short prompt, read out by SIM-o). The two need not respond to the same knob.

We train one iso-parameter width–depth grid of masked-diffusion codec language models and then sweep T at inference only, which yields a full (w, d, T) response surface per metric at zero additional training cost. Four hypotheses were pre-registered before the first run: that refinement is metric-selective (H-D2, primary), that steps and depth exchange at a rate κ (H-D3), that shape matters at fixed parameter count (H-D1), and that small budgets predict large ones (H-D4). Table 7 gives every verdict, including the failures. Eq. (1) states the two fitted forms: one separable (M_{sep}) in width w , depth d and steps T , one (M_{sub}) in which steps substitute for depth through the product dT^κ .

$$\begin{aligned} \text{separable: } \text{err}_m(w, d, T) &= E_m + A_m w^{-\alpha_m} + B_m d^{-\beta_m} + C_m T^{-\tau_m} \\ \text{substitution: } \text{err}_m(w, d, T) &= E_m + A_m w^{-\alpha_m} + B_m (dT^{\kappa_m})^{-\beta_m} \end{aligned} \quad (1)$$

Here E_m is irreducible error, A_m, B_m, C_m are amplitudes, $\alpha_m, \beta_m, \tau_m$ exponents and κ_m the exchange rate. The pre-registered test is supported in the coordinate it declared: $\Delta\tau = 0.110$, run-level bootstrap CI [0.092, 0.131], excluding zero. But τ is only guaranteed invariant to *affine* rescalings of the error, which is what our pre-registration checked, and $\Delta\tau$ changes sign under monotone non-affine ones, $\log(1 + \text{err})$ and $\sqrt{\text{err}}$ among them (§4.3). A quantity whose sign depends on whether one writes WER or $\sqrt{\text{WER}}$ describes a coordinate, not a system. We therefore restate the finding in a form that does not move: against floors we measure, refinement closes 86.2% of the reachable intelligibility range and 46.4% of the reachable identity range. Depth and steps, meanwhile, do not exchange at a rate one could design around: the substitution form fits *worse* than the separable one ($\Delta\text{AICc} = +69.3$ for WER, $+71.3$ for SIM-o), and κ runs to its bound, so no single ratio holds across the grid. Extra inference compute is a targeted instrument, not a substitute for architecture.

Contributions. ❶ The first per-metric (w, d, T) scaling surface for speech generation: 75 points per metric from 45 trained models, released with the training and analysis code. ❷ A floor-referenced, model-free measurement of what refinement buys per capability (86.2% versus 46.4% of the reachable range), together with the scope conditions under which the usual exponent-based statement of the same effect fails: a reparameterisation, a fitted range, and a goodness-of-fit test its own criterion rejects. ❸ A per-axis account of what moves speaker identity, measured against a codec ceiling: training compute and test-time selection move it where refinement does not, while longer prompt context and rate matching buy nothing, with the selection result confirmed by 4 encoder families outside the selector’s lineage.

2 Related work

Masked diffusion language models [Austin et al., 2021, Lou et al., 2024, Sahoo et al., 2024] are trained at scale [Nie et al., 2025a,b], but their scaling is reported against training compute. The masked-generative recipes that made codec-token TTS practical [Borsos et al., 2023, Chang et al., 2022, Wang et al., 2025, Ju et al., 2024] fix a step budget by engineering judgement and ablate it at a few settings; none maps error against steps as a law. Autoregressive codec TTS [Chen et al., 2025] has been scaled in parameters, data and verifier-guided search [Ye et al., 2025], which separates neither width from depth nor either from steps. Classical scaling work aggregates the same way [Kaplan et al., 2020, Hoffmann et al., 2022]. Test-time scaling is well studied in reasoning [Snell et al., 2025], where the payoff is a single correctness metric, and in image diffusion, where search keeps paying after step gains flatten [Ma et al., 2025]. Whether extra inference compute helps different capabilities *differently* is, to our knowledge, untested in generative speech.

3 Method

Backbone and objective. The backbone is a bidirectional (non-causal) pre-LN transformer over Mimi [Défossez et al., 2024] codec tokens at 12.5 Hz, using the first 8 residual codebooks (2048

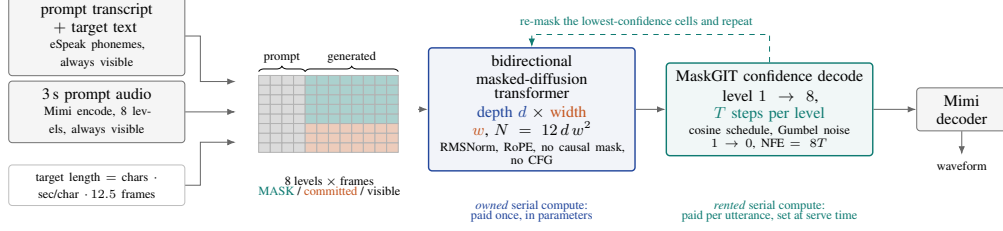


Figure 1: *The two ways a masked-diffusion TTS model buys sequential computation.* Phonemes [Bernard and Titeux, 2021] and a 3-second prompt are always visible; the generated region of the $8 \times \text{frames}$ token grid starts fully **masked**. The backbone spends **depth** (*owned* serial compute, fixed at training time), and the sampler spends **T refinement steps per codebook level** (*rented* serial compute, chosen at serve time). One training grid over (w, d) therefore yields a whole (w, d, T) surface at inference. This paper asks whether the rented kind helps every capability equally.

entries each, plus MASK and PAD). It is parameterised by **width** w and **depth** d , with head dimension 64, GELU MLP ratio 4, RMSNorm, RoPE [Zhang and Sennrich, 2019, Su et al., 2024] and no weight tying, so the non-embedding parameter count is $N = 12 d w^2$. One embedding table per codebook is summed at the input and one linear head per codebook reads out. Training follows SoundStorm-style [Borsos et al., 2023] coarse-to-fine masking: per example we draw a level $l \sim \mathcal{U}\{1..8\}$ and a ratio $t \sim \mathcal{U}(0, 1]$; levels below l stay visible, level l is masked i.i.d. with probability t , and levels above l are entirely MASK. The loss is cross-entropy on the masked cells of level l only. A phoneme prefix and a 3-second prompt of codec tokens are always visible. The model has no speaker encoder, so the backbone is the only identity pathway, and it is trained without classifier-free guidance (CFG; Ho and Salimans, 2022), which would otherwise confound the step axis. The target length is fixed before generation from the text length and a corpus character rate (0.0602 s/char).

Data and task. All models are trained on a 2,000 h speaker-stratified subset of Emilia-EN [He et al., 2024] covering more than 4,000 speakers, with 200 held out entirely. Evaluation is cross-sentence zero-shot synthesis (prompt and target are *different* utterances from the same held-out speaker, so copying the prompt’s content does not score) on 400 items from 174 unseen speakers, targets 4–15 s. Two measured floors anchor every number we report: the word error Whisper-large-v3 [Radford et al., 2023] makes on the real recordings of those items (0.0345), and the speaker similarity of a Mimi round trip (encode, then decode) of the real target with no model in the loop (0.5554, the *codec ceiling*). Neither is reachable by a better generator, and referencing gains to zero instead of to these floors is what makes the two metrics look incomparable.

Frozen sampler. Inference is MaskGIT-style confidence decoding [Chang et al., 2022], run level by level with exactly **T refinement steps per level**, so the number of function evaluations is $\text{NFE} = 8T$. A cosine schedule sets how many cells remain masked after each step, tokens are drawn at temperature 1.0, and confidences are log-probabilities plus Gumbel noise annealed $1 \rightarrow 0$; committed cells stay committed. The sampler is identical for every config, seed and T , and generation noise is keyed so that T is the only difference between surface points of one run.

Design and statistics. The grid holds 15 configs: three iso- N budgets $\times$ five aspect ratios, each trained with 3 seeds for 30,000 steps at effective batch 256 with maximal-update parameterisation (μP) [Yang et al., 2021] (base width 256, base LR 0.004, transfer checked by five-point LR sweeps). We sweep $T \in \{1, 2, 4, 8, 16\}$ at evaluation only, which gives 75 surface points per metric from 45 trained models. The metric stack is WER (Whisper-large-v3 [Radford et al., 2023], greedy), SIM-o (WavLM-large [Chen et al., 2022] cosine against the *original* prompt waveform, following Le et al. [2023], Anastassiou et al. [2024]), a frozen degenerate-output rule (transcript empty, under 30% of the reference length, or a 4-gram repeated 6 $\times$), and UTMOS [Saeki et al., 2022] for predicted naturalness. The error transforms are $\text{err}_{\text{WER}} = \text{WER}$ and $\text{err}_{\text{SIM}} = 1 - \text{SIM-o}$, computed over all items including degenerates. We fit both forms of Eq. (1) by weighted non-linear least squares with 32 multi-starts. The intervals on $\Delta\tau = \tau_{\text{WER}} - \tau_{\text{SIM}}$, κ and $\Delta\rho = \alpha_{\text{SIM}}/\beta_{\text{SIM}} - \alpha_{\text{WER}}/\beta_{\text{WER}}$ are 95% percentile intervals from a bootstrap [Efron, 1979] of 2000 replicates that resamples at the *run* level and recomputes every T point. The run level is the right unit because the five T points of one run share a trained model, so resampling surface points would understate their correlation.

4 Results

4.1 What refinement buys, measured against floors

The comparison that matters is not how much error each metric sheds (WER and 1–SIM-o have no common unit) but how much of the error that is *reachable at all* each one sheds. Table 1 reports that fraction directly from the measurements, with no fitted model. Going from one-step decoding to the deployable $T=16$ default, refinement closes 86.2% (CI [82.8, 89.1]) of the distance from $T=1$ to the automatic speech recognition (ASR) floor, and 46.4% (CI [45.0, 47.8]) of the distance to the codec ceiling: a ratio of $1.86\times$ (CI [1.82, 1.90]). Refinement very nearly exhausts what is available to it on intelligibility while leaving most of the identity range untouched (Fig. 4).

Two properties make this the statement we build on. First, it is stable in T : we extended every run to $T=64$, and over that $16\times$ range the ratio stays between $1.85\times$ and $1.96\times$ while $\Delta\tau$ halves (§4.3). Quadrupling the budget past the deployable default takes intelligibility from 86.2% to 88.4% but identity only from 46.4% to 47.9% (CI [46.6, 49.2]), and pushing the largest budget to $T=128$ moves WER by +0.0018 and SIM-o by -0.0052: the refinement axis has stopped paying and begun, slightly, to cost. Second, it is stable in the coordinate: because it is a ratio of differences along one axis, a monotone reparameterisation moves numerator and denominator together, so across $\log(1 + \text{err})$, $\sqrt{\text{err}}$ and err^2 the ratio stays between $1.68\times$ and $1.86\times$ where $\Delta\tau$ changes sign.

4.2 The two capabilities have different bottlenecks

The asymmetry is not an intrinsic property of the two capabilities, and saying so is the paper’s main claim rather than a caveat on it. We retrained the same 9 runs (configs C1/C3/C5, i.e. the shallowest, middle and deepest 125 M shapes, $\times 3$ seeds, identical at every budget) for $3\times$ and $6\times$ the original schedule and repeated the measurement (Table 3). The ratio attenuates monotonically ($1.84\times$, $1.36\times$, $1.23\times$) with non-overlapping intervals, and stays strictly above one throughout.

What drives it is visible in the two shares separately. *Refinement’s intelligibility share saturates and stays saturated*: across the $2\times$ step from 90k to 180k it moves by -0.09 percentage points, and a second recogniser from a different family (wav2vec2-large-960h-lv60-self [Baevski et al., 2020, Xu et al., 2021], a connectionist temporal classification (CTC) model) puts the same figure at -0.06 points: two systems that disagree by a factor of 2.4 on absolute word error agree to a tenth of a point that further training buys no intelligibility. At 180k our models reach that recogniser’s word error on the *real recordings* to within 0.1501 absolute, so there is nothing left to close. *The identity share, over the same step, rises by +7.4 points and is still rising*.

The selectivity therefore measures how far from converged each capability is, not what refinement is able to buy. Intelligibility is bounded by refinement and reaches that bound early; identity is bounded by training compute and by the codec, and refinement cannot reach either. This is why the ordering, and not the factor, is what we report: the factor is a property of the training budget, and under-training inflates it. Three budgets do not determine whether the ratio converges above one or to one, and we do not extrapolate.

The grid resolves shape differences well above seed noise: within each budget the spread of config means at $T=16$ is at least twice the pooled seed standard deviation, so the pre-registered power criterion passed in all three budgets.

4.3 The same effect as a fitted exponent, and why we do not lead with it

Our pre-registration declared a power-law surface (Eq. 1) and a primary test on its step exponents. That test passes: $\tau_{\text{WER}} = 0.836$ (CI [0.825, 0.851]) against $\tau_{\text{SIM}} = 0.726$ (CI [0.709, 0.744]), giving $\Delta\tau = 0.110$, CI [0.092, 0.131], excluding zero (H-D2). Three scope conditions bound what it means, and we state them because each one weakens a claim we would otherwise have made.

❶ *Coordinate*. τ is invariant to affine rescalings of the error, which is what we verified before running; it is not invariant to monotone non-affine ones. Refitting the identical measurements under $\log(1 + \text{err})$, $\sqrt{\text{err}}$ and a cap at $\text{err} = 1$ sends $\Delta\tau$ *negative* in 3 of the reparameterisations we tried (Table 2). Its sign is a fact about the coordinate. ❷ *Range*. τ summarises curvature over the window it is fitted on, and is not an identified asymptotic exponent. If we extend every run to $T=64$ and refit on nested windows, $\Delta\tau$ falls from +0.2191 over $T \leq 8$ to +0.1102 over the declared $T \leq 16$

and +0.0660 over $T \leq 64$: it roughly halves each time the window doubles. The declared value is a local slope at the edge of the declared grid. **Fit.** Judged by the weighted criterion the fit itself minimises, the declared form is rejected as a generative model: $\chi^2/\text{dof} = 14.3$ for WER and 8.8 for SIM-o. We use it as a compact description of curvature and read no mechanism from its parameters. The declared interval also holds the $1/\text{SE}^2$ weights (SE: standard error over seeds) fixed at their observed-data values; resampling the weights as well widens it to [0.0349, 0.1640].

None of this touches §4.1, which is model-free. It is why §4.1 is the result we lead with.

Table 1: *What refinement buys, as a fraction of what is reachable.* Share of the distance from one-step decoding to a measured reference, Whisper’s word error on the real recordings (0.0345) and the speaker similarity of our codec’s round trip (0.5554), closed by step budget T , on the full 45-run grid. Percentile CIs from a 2000-replicate run-level bootstrap. No model is fitted, and the ratio is invariant to monotone reparameterisation of the error ($1.68\times\text{--}1.86\times$, Table 2).

T	fraction of reachable range closed		
	intelligibility (WER)	identity (SIM-o)	ratio
2	32.2% [29.7, 34.4]	20.0% [19.1, 20.9]	$1.61\times$
4	69.8% [66.0, 73.1]	35.7% [34.3, 37.0]	$1.96\times$
8	81.8% [78.3, 84.8]	43.5% [42.1, 44.9]	$1.88\times$
16	86.2% [82.8, 89.1]	46.4% [45.0, 47.8]	$1.86\times$
32	87.9% [84.9, 90.5]	47.5% [46.2, 48.8]	$1.85\times$
64	88.4% [85.3, 91.0]	47.9% [46.6, 49.2]	$1.85\times$

Table 2: *The exponent contrast is a property of the error coordinate.* The same 75 measurements, refitted after a monotone reparameterisation of both errors. τ is invariant to the affine map, as our original (v1.0) pre-registration assumed; under the non-affine maps $\Delta\tau$ changes sign, while the model-free ratio of Table 1 does not. This is why the paper’s claim is stated as a share of a reachable range rather than as an exponent difference.

error coordinate	τ_{WER}	τ_{SIM}	$\Delta\tau$	gap-closed ratio
identity (<i>pre-registered</i>)	0.8362	0.7260	+0.1102	$1.86\times$
$\log(1+\text{err})$	0.6665	0.6913	-0.0248	$1.85\times$
$\sqrt{\text{err}}$	0.4325	0.6842	-0.2517	$1.68\times$
err capped at 1	0.7093	0.7260	-0.0167	n/a
err squared	1.5613	0.8122	+0.7491	$1.82\times$
affine 3+2*err (v1.0 check)	0.7600	0.7249	+0.0351	n/a

Table 3: *Refinement’s intelligibility share saturates; its identity share does not.* The same 9 runs ($\text{C1/C3/C5} \times 3$ seeds) retrained to three budgets and measured identically: share of the reachable range closed from $T=1$ to $T=16$. The second intelligibility column re-transcribes the identical audio with wav2vec2-large-960h-lv60-self, a CTC recogniser sharing neither architecture, objective nor training corpus with Whisper; it is reported so that saturation cannot be an artefact of one recogniser. Intervals cluster the bootstrap on *configuration* (seeds within a config are not draws from config space, so the effective n is 3) and resample evaluation items as a second dimension.

training	intelligibility share		identity		
	Whisper	wav2vec2-large-960h-lv60-self	share	ratio	95% CI
30k	90.6%	88.5%	49.3%	$1.84\times$	1.73, 1.97
90k	97.1%	99.2%	71.5%	$1.36\times$	1.28, 1.44
180k	97.0%	99.1%	78.8%	$1.23\times$	1.18, 1.29

4.4 Robustness of the asymmetry

The selectivity is not an artefact of scale, training compute or the metric stack (Table 12). A fourth parameter budget (276 M, ten further runs) gives $\Delta\tau = +0.1095$, CI [0.0951, 0.1241]; retraining three configurations $3\times$ longer gives +0.1767, CI [0.0789, 0.2854]; a different ASR gives +0.1283 and unbounded log-amplitudes +0.1260. Extending T to 64 leaves WER still falling past the grid edge ($0.1516 \rightarrow 0.1294$), so the gain measured at $T=16$ is a lower bound.

Depth and steps do not substitute. The substitution form of Eq. (1) fits the surface worse than the separable form by the corrected Akaike information criterion: $\Delta\text{AICc}(M_{\text{sub}} - M_{\text{sep}}) = +69.3$ for WER and $+71.3$ for SIM-o, where the pre-registered claim (H-D3) required $\leq +4$, and κ sits at its bound in every fit. The honest reading is narrower than “no exchange rate exists”. A global factor dT^κ is a strong constraint: it demands that one ratio hold at every depth and every step count, and it is that constraint the data reject (Fig. 2, right). The separable fit still implies a *local* rate, $\kappa \approx \tau/\beta = 0.41$ for WER, valid near the operating point and nowhere claimed to extend beyond it. What does not survive is the design rule: one cannot trade depth for steps at a fixed rate across the grid.

Shape matters, but the anisotropy ratio is not identified. At $T=16$ the two-axis model beats the N -only model for both metrics ($\Delta\text{AICc} = -15.5 / -5.5$ against a -4 threshold; Hurvich and Tsai, 1989), so error at fixed parameter count depends on how those parameters are shaped (Fig. 2, left). The direction of that dependence does not resolve. The width amplitude saturates its pre-registered upper bound in every fit, so only the product $A w^{-\alpha}$ is identified and α is not, which leaves the anisotropy ratio $\rho = \alpha/\beta$ undetermined and the secondary pre-registered claim (H-D1) not supported.

Depth is identified, and it separates the capabilities sharply ($\beta_{\text{WER}} = 2.040$ versus $\beta_{\text{SIM}} = 0.553$; Table 11): moving from the shallowest shape to the best depth is worth 40.2, 16.3 and 7.3 WER points across the three budgets, while SIM-o varies by under 0.065. The depth optimum is interior only in the smallest budget; in the two larger ones the argmin sits at the deepest shape tested, so those are lower bounds, not interior optima.

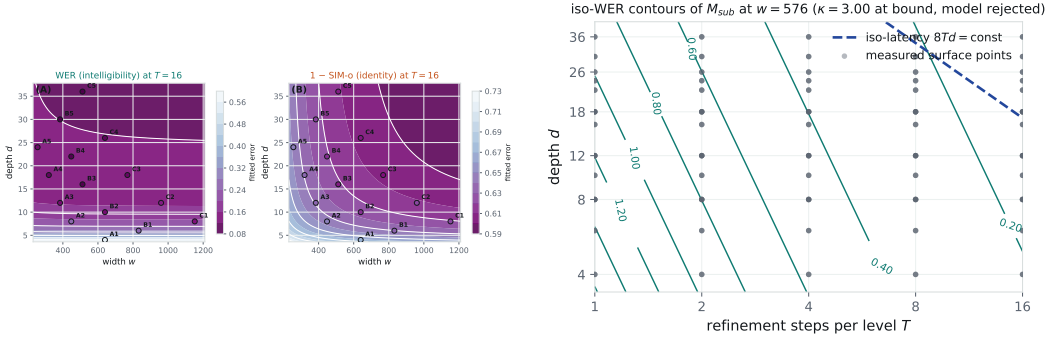


Figure 2: *Left*: iso-error contours in (w, d) at $T=16$. The two capabilities do not share a preferred shape, but the width amplitude saturates its bound, so the anisotropy ratio ρ is not identified. *Right*: iso-WER contours in $(\log T, \log d)$: a single substitution factor dT^κ cannot reproduce both the depth ordering at fixed T and the T ordering at fixed depth.

Extrapolation and degenerate outputs. Fits on the two smaller budgets do not predict the largest (H-D4 refuted), and the frozen degenerate rule fires on 0.1% of items at $T=16$, concentrated at the shallowest shapes. Both are reported in full in the appendix (Fig. 5, Appendix B).

Serving corollary. Per-layer latency is flat in width: measured at batch 1, c_{layer} is 0.493–0.514 ms for $w \in [256, 1152]$, because the device is launch-bound at this scale. Serial latency is therefore $8Td c_{\text{layer}}$, so at fixed parameter count width is free and a latency budget constrains only the product Td . How to divide that product between depth and steps is not determined here. The pre-registered rule, buy depth before steps, is refuted: it holds at 5.9% of the 40 latency budgets we test, against a required 80%. The reverse ordering is a boundary effect of a grid censored at $T=16$, not a finding.

Training compute is the largest identity lever, and has not saturated. Retraining three configurations to 90k steps raises SIM-o by $+0.0827$ (CI [0.0733, 0.0908], 9 runs, all positive), and a further doubling to 180k adds $+0.0155$ (CI [0.0064, 0.0243]) at a WER cost inside the ± 2.0 -point guardrail. Across $\{30k, 90k, 180k\}$ steps on one configuration, SIM-o runs $0.4081 \rightarrow 0.4807 \rightarrow 0.4962$. Extrapolating the first interval log-linearly predicts 0.5264 at 180k against 0.4962 observed, a residual of -0.0302 : the axis is decelerating but still paying, and per unit it buys more identity than any inference-time intervention we tested.

Table 4: *What buys speaker identity.* Absolute SIM-o gain per axis, set against the 0.0747 of headroom between our best system and a codec round trip of real audio. The headroom is a WavLM-large quantity, so every row is reported under that same encoder (final column); the cross-encoder agreement evidence for the search row is separated out into Table 6 rather than mixed into this column. Training compute and test-time selection buy the most. Guidance buys a little and charges for it in WER; rate matching and longer prompts buy nothing.

axis	change	Δ SIM-o	cost	scorer
training compute	30k $\rightarrow$ 90k steps	+0.0827	3 $\times$ training	WavLM-L
(further)	90k $\rightarrow$ 180k	+0.0155	2 $\times$ again	WavLM-L
test-time search	best-of-8 vs $T=16$	+0.0365	+0.0425 WER, 8 $\times$ NFE	WavLM-L
parameters	20 M $\rightarrow$ 276 M	+0.0531	14 $\times$ parameters	WavLM-L
guidance	$\gamma = 0.5$	+0.0087	+4.41 WER points	WavLM-L
rate matching	per-item duration	+0.0044	free	WavLM-L
prompt context	3 s $\rightarrow$ 9 s	none	+71.7 WER points	WavLM-L

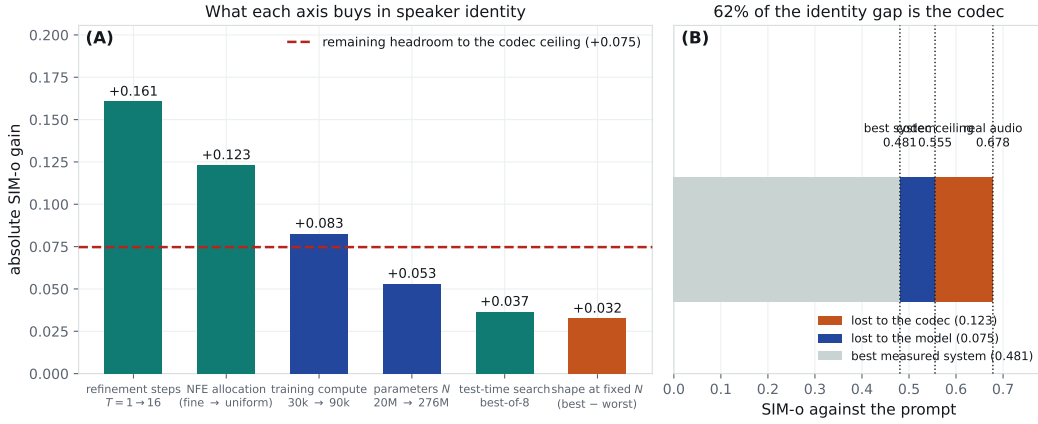


Figure 3: *Identity is mostly not the model's to win.* (A) what each axis buys in speaker similarity, against the headroom that remains to the codec ceiling; bars are coloured by which lever they are (rented at serve time, owned in training, shape). Refinement's bar is the largest only because it is measured from one-step decoding, which is not an operating point anyone ships; from the deployable $T=16$ default it buys nothing further, and test-time search is what moves identity from there. (B) of the distance between real audio and our best system, 62% is lost in the codec itself.

What buys identity, measured against a ceiling. If refinement is an intelligibility knob, what moves identity? We put every candidate lever on one scale (Table 4, Fig. 3) and read it against a fixed bound. A codec round trip of real audio scores 0.555, leaving 0.0747 of headroom above our best system.

Training compute is the largest lever: 30k $\rightarrow$ 90k steps gains +0.0827 (CI [0.0733, 0.0908] over 9 runs), and a further doubling adds +0.0155. Test-time search is the second lever: best-of- K sampling at matched NFE ($8T$ per candidate), selecting with one speaker encoder and scoring with a second. It gains +0.0444 over refinement at $T=64$ (CI [0.0416, 0.0470], on 72% of items) and +0.0365 over the deployable $T=16$ default, at a cost of +0.0425 WER. Refinement remains the better spend for intelligibility: WER is 0.1316 under refinement against 0.1955 under search. The gain comes from selection, not from sampling: an arbitrary candidate scores 0.4259 regardless of K . It persists at 276 M non-embedding parameters (+0.0422) and, on three runs, at 3 $\times$ training compute (+0.0368).

The remaining levers are small or empty. A 14 $\times$ parameter increase gains +0.0531. Training-free speaker-contrastive guidance (logits pushed away from those under a wrong speaker's prompt, with weight γ) gains +0.0087 and costs +4.41 WER points, so it trades one capability for the other (Table 8). Rate-matched length conditioning gains +0.0044, and longer prompts gain nothing (Table 9): the best prompt is the 3 s default seen in training, and beyond it the output stays fluent and speaker-consistent while decoupling from the text. Identity is bought with training and with selection, not with more refinement.

Where the passes are spent matters as much as how many: at matched NFE = 32, moving them to the coarsest codebook level changes WER from 1.1480 to 0.2191, at a cost in identity and naturalness (Appendix D).

Table 5: *Refinement and search buy different capabilities at the same cost.* Matched NFE (8T per candidate); best-of- K candidates are selected by WavLM-SV and scored by a separate ECAPA encoder. Refinement wins intelligibility at every tier, and search wins identity at every tier.

NFE	refinement	search	Δ ECAPA	95% CI	WER (refine / search)
128	$T=16$	$K=2$	+0.0153	[0.0127, 0.0179]	0.1530 / 0.1938
256	$T=32$	$K=4$	+0.0322	[0.0307, 0.0338]	0.1335 / 0.1947
512	$T=64$	$K=8$	+0.0444	[0.0416, 0.0470]	0.1316 / 0.1955
replicated at NFE 512 outside the grid: 276 M budget +0.0422 [0.0393, 0.0448]; $3\times$ training +0.0368 [0.0321, 0.0428]					

Table 6: *The search result is not an artefact of the selector’s lineage.* The same 200 items per run, 15 runs, rescored end-to-end by five speaker encoders spanning four model families. The selector is WavLM-large SV, so its own row is circular and is shown only for reference. A *per-item win rate* (how often an encoder ranks the search output above the refinement output at matched NFE) is invariant to any monotone rescaling of that encoder’s cosine, which a difference of means is not; we therefore compare win rates across encoders and report each row Δ only beside its own scale. Every family, including the three whose absolute cosines fail our ground-truth gate (median same-speaker cosine ≥ 0.50 , cross-speaker ≤ 0.25) *purely on scale* while separating speakers at AUC ≥ 0.98 , puts search ahead with an interval clear of one half.

encoder	family / role	win rate	95% CI	raw Δ
ECAPA-TDNN [Desplanques et al., 2020]	ECAPA, gated	72.3%	[70.6, 73.8]	+0.0444
x-vector [Snyder et al., 2018]	TDNN	79.0%	[77.5, 80.5]	+0.0075
GE2E d-vector [Wan et al., 2018]	LSTM d-vector	69.1%	[67.5, 70.8]	+0.0164
WavLM-base+ SV	WavLM, other scale	64.6%	[62.9, 66.3]	+0.0173
WavLM-large SV*	WavLM / selector	84.9%	[83.6, 86.1]	+0.0803
*selects the candidates, so this row is circular and is not counted as agreement.				

5 Limitations and outlook

Every model in the grid is under-trained. We cap non-embedding parameters at 133 M and training at 30,000 steps, so WER’s saturation step T_{WER}^* is a lower bound. Part of the identity gap is also not ours to close: a codec round trip of the target scores 0.555 against 0.678 for the original waveform, so 62% of the distance to real speech is lost in the codec before the model generates anything.

Our selector and our original scorer share a lineage (both are ECAPA-TDNN-family models, correlating per item at r in the range 0.71–0.73), so the search result was open to the charge that it measured a shared representational bias rather than identity. Table 6 closes that charge but does not make the measurement lineage-free: the candidates are still *chosen* by one encoder, and what the other four establish is that the chosen candidates are better by their lights too, not that a different selector would choose the same ones. Compute matching (Table 5) also covers the diffusion transformer only, so the selector and scorer forward passes are not charged to the search arm.

The study covers one language, one codec and one corpus, with no classifier-free guidance. Target length comes from a corpus median character rate rather than a per-item prediction. Two fitted quantities are not identified: κ sits on its bound in every fit, and the width amplitude saturates its bound, which leaves $\Delta\rho$ undetermined. The $\Delta\tau$ interval reflects seed variation only, and widens to [0.0349, 0.1640] once the fit’s weights are resampled along with the runs.

Table 7 in Appendix B gives every pre-registered verdict. Four extension (H-E) and speaker-identity (H-S) hypotheses were refuted outright (H-E1b, H-E2, H-E4b, H-S4), three were not supported (H-D1, H-D3, H-D4) and two were discordant between their pre-specified lenses (H-S1, H-S3); we name them because a paper that quotes only the passing half of a split hypothesis cannot be audited. Two censorings remain, both removable by a larger grid: the depth optimum sits at the deepest shape tested in the two larger budgets, and WER is still falling at the top of the fitted T range.

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A Reproducibility

Protocol, hypotheses, thresholds, the frozen sampler and the analysis plan were registered before the first run, and every gate is logged with its measured values. The released artefact holds the harness, all 45 run records, the 75-point surface, and the fitting and bootstrap code. Training the grid took 162.5 GPU-hours.

B The shape grid, as run

Every config in the active grid with its measured $T=16$ means (seeds 0, 1, 2, 400 items each from the zero-shot evaluation set eval_zs). Iso- N within a budget: the three budgets (A, B, C) sit at 20, 50 and 125 M non-embedding parameters, and the five shapes per budget span the aspect-ratio range at fixed $N = 12dw^2$.

config	budget	w	d	heads	N (M)	val loss	WER (%)	SIM-o	Degen (%)
A1	A	640	4	10	19.7	4.638	53.9	0.327	0.67
A2	A	448	8	7	19.3	4.582	28.4	0.348	0.08
A3	A	384	12	6	21.2	4.566	15.9	0.360	0.17
A4	A	320	18	5	22.1	4.574	13.6	0.359	0.00
A5	A	256	24	4	18.9	4.611	15.3	0.351	0.17
B1	B	832	6	13	49.8	4.551	27.8	0.361	0.08
B2	B	640	10	10	49.2	4.505	16.0	0.381	0.00
B3	B	512	16	8	50.3	4.497	11.9	0.388	0.08
B4	B	448	22	7	53.0	4.511	12.0	0.382	0.08
B5	B	384	30	6	53.1	4.520	11.5	0.375	0.08
C1	C	1152	8	18	127.4	4.504	17.1	0.373	0.17
C2	C	960	12	15	132.7	4.504	18.9	0.373	0.17
C3	C	768	18	12	127.4	4.488	14.3	0.392	0.00
C4	C	640	26	10	127.8	4.489	13.4	0.390	0.08
C5	C	512	36	8	113.2	4.487	9.9	0.388	0.00

Table 7: *The complete pre-registration ledger.* Every hypothesis declared before the first run (H-D: main grid, v1.0; H-E: extensions; H-S: speaker identity), with its verdict, including the four that failed and the two that split. We report the failures beside the successes because a ledger that shows only its winners cannot be used to judge the winners. DISCORDANT is a declared outcome class, used where the two pre-specified lenses (primary and secondary tests) disagree, and is not a synonym for refuted.

id	claim	verdict
H-D1	fixed- T anisotropy $\Delta\rho > 0$	not supported
H-D2	$\Delta\tau > 0$, metric-selective steps	supported [§]
H-D3	depth-step exchange rate κ	not supported
H-D4	small budgets predict the largest	not supported
H-E1a	$T_{\text{WER}}^* > 16$	supported
H-E1b	extended τ inside the v1.0 CI	refuted
H-E2	iso-latency allocation is depth-first	refuted
H-E3	coarse-level NFE beats fine	supported
H-E4a	scale persistence, four budgets	supported
H-E4b	predicted vs observed within 15 %	refuted
H-E5	survives $3\times$ training compute	supported
H-S1	longer prompt context buys identity	discordant
H-S2	search buys identity	supported
H-S3	rate-matched length buys identity	discordant
H-S4	contrastive guidance buys identity	refuted

[§]supported as declared; see §4.3 for the coordinate, range and fit conditions that bound it.

Table 8: *Inference-time interventions that do not buy identity.* Speaker-contrastive guidance trades intelligibility for similarity and degrades monotonically in γ ; per-item rate matching gives a small mean gain that is not reliable item-by-item.

intervention	setting	Δ SIM-o	95% CI	Δ WER (pts)
contrastive guidance	$\gamma = 0.5$	+0.0087	[0.0063, 0.0115]	+4.41
	$\gamma = 1.0$	-0.0021	[-0.0059, 0.0023]	+8.94
	$\gamma = 2.0$	-0.0322	[-0.0367, -0.0274]	+19.10
rate matching	per-item duration	+0.0044	[0.0017, 0.0067]	+0.86
13/15 runs positive, but a per-item sign test gives $p = 0.051$				

Table 9: *Longer prompts do not buy identity, and destroy intelligibility.* Prompt length swept at inference on 15 runs of the 125 M budget, 397 items, every arm scored against the same fixed 3 s reference. Training used 3 s prompts only, so the 6 and 9 s arms test extrapolation. There the output stays fluent and speaker-consistent, but it decouples from the conditioning text.

prompt	SIM-o	WER
1.5 s	0.3213	0.1578
3 s (<i>training length</i>)	0.3839	0.1474
6 s	0.3724	0.7692
9 s	0.3759	0.8642

Table 10: *Where the refinement passes are spent matters as much as how many.* Three frozen per-level schedules (steps per codebook level, coarsest first) at matched total NFE = 32 on the 125 M budget, 15 runs, 400 items. Spending every pass on the coarsest codebook level rather than the finest cuts WER by $5.2\times$, and costs identity and naturalness against a uniform split.

schedule	WER	SIM-o	UTMOS	degenerate
coarse [25, 1, ..., 1]	0.2191	0.3057	2.17	0.1%
uniform [4, ..., 4]	0.3144	0.3477	2.51	0.5%
fine [1, ..., 1, 25]	1.1480	0.2246	1.56	39.1%

Table 11: *Fitted exponents* on the 75-point surface (M_{sep} for α, β, τ ; M_{sub} for κ), with 95% run-level bootstrap intervals on the step exponents. The primary result is $\Delta\tau = 0.110$, CI [0.092, 0.131], which excludes zero. κ sits on its bound in both rows, and the substitution form is worse than the separable one by $\Delta\text{AIC}_{\text{csub-sep}} \gg +4$, so the depth-step exchange rate has no finite estimate. The negative $\Delta\text{AIC}_{\text{cfull-N}}$ (M_{full} , the separable form without its T term, fitted at $T=16$, against a single power law in N) says that shape matters at fixed N .

Metric	α (width)	β (depth)	τ (steps)	κ	$\Delta\text{AIC}_{\text{csub-sep}}$	$\Delta\text{AIC}_{\text{cfull-N}}$
WER	1.574	2.040	0.836 [0.825, 0.851]	3.000 [†]	+69.3	−15.5
SIM-o	0.919	0.553	0.726 [0.709, 0.744]	3.000 [†]	+71.3	−5.5

[†]at the fitted bound, i.e. not identified; $\Delta\tau = 0.110$, CI [0.092, 0.131].

Table 12: $\Delta\tau$ survives variation in scale, training compute and the measurement stack. Each row refits the surface with one factor changed: a fourth parameter budget, $3\times$ training compute, a different ASR, and an unbounded reparameterisation of the fitted amplitudes. Every interval excludes zero.

variation	what changes	$\Delta\tau$	95% CI
none (75 points, 45 runs)	n/a	0.110	[0.092, 0.131]
fourth budget, 276 M	10 further runs	+0.1095	[0.0951, 0.1241]
$3\times$ training compute	3 configs at 90k steps	+0.1767	[0.0789, 0.2854]
ASR swapped	Whisper-medium.en	+0.1283	[0.1064, 0.1492]
unbounded amplitudes	$\log A, \log B, \log C$ free	+0.1260	[0.1043, 0.1474]

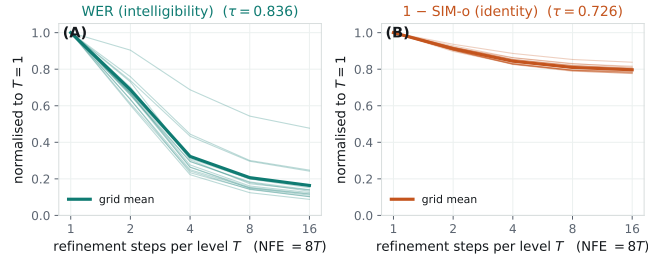


Figure 4: *Test-time scaling is metric-selective*. Error versus refinement steps for all 15 configs, each normalised to its own $T=1$ value; thick lines are grid means. (A) WER error falls 0.952 in absolute terms over $T=1 \rightarrow 16$ and is still falling at $T=16$. (B) SIM-o error falls 0.161 over the same range, $5.9\times$ less. Same models, same frozen sampler, same items; only T differs.

C Additional figures

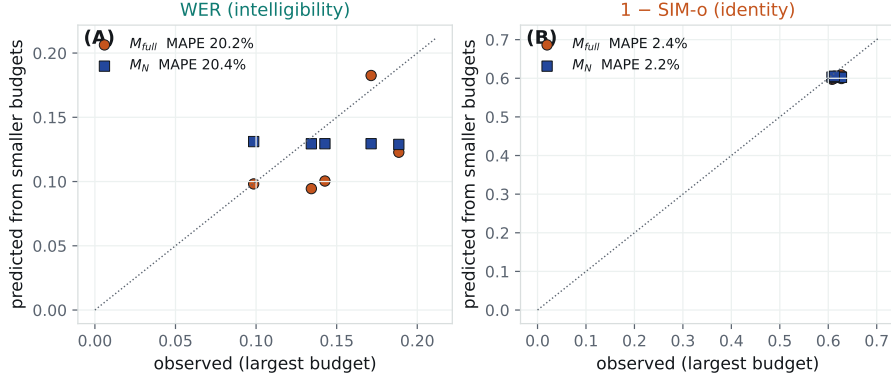


Figure 5: *H-D4 extrapolation*. The two smaller budgets are fitted and the largest budget predicted, under M_{full} and the N -only model. SIM-o extrapolates (mean absolute percentage error, MAPE, 2.4 % vs 2.2 %); WER does not (20.2 % vs 20.4 %).

D Refinement-budget allocation across codebook levels

Three frozen per-level step schedules at matched total NFE = 32 on the 125 M budget (15 runs, 400 items). Paired item-level bootstrap, 2000 replicates, pairing on the eval item so item difficulty cancels. Coarse spends all passes on codebook level 0 (the coarsest), fine on level 7 (the finest), and uniform splits evenly.

schedule	WER	SIM-o	UTMOS	degenerate
coarse [25, 1, ..., 1]	0.2191	0.3057	2.17	0.1%
uniform [4, ..., 4]	0.3144	0.3477	2.51	0.5%
fine [1, ..., 1, 25]	1.1480	0.2246	1.56	39.1%

Allocation is metric-selective in the same way the step *count* is: coarse wins WER but loses identity and naturalness against uniform, both with intervals excluding zero. It is a trade, not a free win.

A significance-based saturation contrast. Defining saturation as the smallest T whose paired-bootstrap interval for $\text{err}(T) - \text{err}(64)$ covers zero gives $T_{\text{WER}}^* = 32$ against $T_{\text{SIM}}^* = 16$ on the extended grid. We report this only here and do not build on it: roughly half the contrast is a precision artefact, since identity’s detection floor is $\approx 1.8\times$ looser relative to its own dynamic range than intelligibility’s. The raw-scale T^* (95% of the $T=16$ value) is not affine-invariant and must never be compared across metrics: the same 5% band is 0.67% of WER’s range but 11.96% of SIM-o’s. On the invariant fraction-of-total-gain definition, $T^* = 16$ for both metrics.

E Fit diagnostics and exploratory analyses

Parameters at bounds. The width amplitude A saturates the pre-registered *upper* bound $A \leq 10$ in every fit (Part A, the $T=16$ width–depth fit: 10.000 for both metrics; Part B, the full surface fit: 9.999 for WER, 10.000 for SIM-o), so only the product $A w^{-\alpha}$ is identified and α is not; the bound, not the data, is binding, which is a limitation of the pre-registered parameterisation worth fixing in any follow-up. κ reaches its upper bound in every one of the 2000 bootstrap replicates of M_{sub} . The irreducible term E reaches 0 for WER under M_{sep} . The depth amplitude B is interior throughout, so β is identified.

Exploratory: the depth exponent separates the capabilities. $\beta_{\text{WER}} = 2.040$ against $\beta_{\text{SIM}} = 0.553$, a factor of 3.7: the asymmetry H-D1 sought, in the parameter the data pin down. Part-B ρ values point the pre-registered way ($\Delta\rho = +0.890$) but inherit the A -cap problem; the pre-registered test is Part A, so this is exploratory only.

Weighting. Points are weighted $1/SE^2$ with SE taken over seeds, floored at a quarter of the pooled seed SD so that a chance agreement between two seeds cannot dominate the fit; the number of floored points is recorded in the released `fits.json`.

Exploratory (not part of any declared claim). An unweighted refit of both parts is released alongside the weighted fit, as is a fit that additionally includes the finished runs of a stretch third seed. Both reproduce the sign and significance of $\Delta\tau$.

Use of Generative AI Disclosure

Generative AI was used as a coding assistant for the experimental harness and for language refinement of author-written text. The research design, the pre-registered hypotheses and thresholds, the analysis plan, and all scientific claims are the authors' own; the authors verified the code and the numbers and take full responsibility for this paper.